Project Synopsis

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1. Title

EchoMateLite: Design and Deployment of a Cloud-Based Lightweight Social Media Platform on Amazon Web Services (AWS)

2. Problem Statement

In today's digitally interconnected world, social media platforms play a central role in human communication, information sharing, and online community building. However, most existing social media solutions are highly complex, resource-intensive, and difficult to scale — making them unsuitable as reference implementations for cloud-based learning or small-scale deployments. There is a clear need for a lightweight, cloud-deployable social media platform that demonstrates core functionalities such as user authentication, profile management, and post creation and viewing, while remaining cost-efficient and scalable.

EchoMateLite addresses this gap by offering a simplified yet fully functional social media platform built using the MERN stack (MongoDB, Express.js, React.js, Node.js). The core challenge lies in architecting a system capable of handling concurrent user sessions, securely storing user-generated content, and serving a responsive web front end — all deployed-on AWS cloud infrastructure. Deploying such a platform on AWS introduces real-world complexity around service selection, security hardening, cost optimization, and infrastructure management. This project directly addresses these challenges by designing and deploying EchoMateLite on AWS, providing practical, end-to-end exposure to cloud application development and deployment workflows aligned with industry standards.

3. Goals and Objectives

Overall Goal:

To design, develop, and successfully deploy a lightweight cloud-based social media platform — EchoMateLite — on Amazon Web Services (AWS), demonstrating practical proficiency in full-stack MERN development, AWS cloud architecture design, and end-to-end cloud deployment.

Specific Objectives:

- To design and develop a fully functional MERN-stack web application (EchoMateLite) that supports user registration, login, profile management, post creation, and a dynamic news feed.

- To design a scalable, cost-efficient AWS cloud architecture capable of hosting the EchoMateLite application with high availability.
- To integrate relevant AWS services — including EC2, S3, RDS/MongoDB, Amazon Cognito, CloudFront, and ELB — to form a complete cloud solution.
- To implement fundamental security practices including HTTPS enforcement, secure password hashing, and IAM-based access control.
- To evaluate the performance, cost, and scalability of the deployed AWS architecture and document findings with architecture diagrams and cost analysis.

4. Key Features / Expected Results

4.1 User Authentication and Account Management

EchoMateLite will implement a secure user authentication system allowing users to register new accounts and log in with credentials. Passwords will be hashed using bcrypt before storage. Amazon Cognito will be integrated to manage user pools, sign-up/sign-in flows, and JWT-based session tokens. Each authenticated request to the backend API will be validated against the issued token. The expected outcome is a fully functional, secure registration and login system with persistent session management.

4.2 Profile Management

Registered users will be able to create and edit their profiles, including their full name, bio, and profile picture. Profile images will be uploaded to and served from Amazon S3, providing scalable and cost-effective object storage. Updates will reflect immediately upon saving. The expected outcome is a dynamic user profile page with real-time data updates.

4.3 Post Creation and News Feed

Users will be able to compose and publish text-based posts visible across the platform. A news feed will aggregate posts from all users and display them in reverse chronological order with pagination support for performance. The backend API will expose RESTful endpoints for creating and fetching posts, with data stored in MongoDB. The expected result is a responsive, real-time feed providing a genuine social media browsing experience.

4.4 Responsive React.js Frontend

The client-side application will be developed using React.js as a Single Page Application (SPA), ensuring fast navigation without full-page reloads. The interface will be responsive and compatible with desktop and mobile devices. The production build will be hosted on Amazon S3 and delivered via Amazon CloudFront for low-latency global access. The expected outcome is a modern, intuitive, and responsive user interface.

4.5 Cloud Deployment on AWS

The entire application stack will be deployed on AWS. The Node.js/Express.js API server will run on an Amazon EC2 instance (t2.micro, Free Tier eligible). The React.js frontend build will be hosted on S3 and served through CloudFront. The database layer will use MongoDB Atlas or Amazon RDS, depending on final evaluation. An Application Load Balancer (ALB) with AWS Certificate Manager (ACM) will enforce HTTPS. The expected result is a live, publicly accessible deployment of EchoMateLite accessible via a secure web URL.

4.6 Security and Best Practices (Optional Enhancement)

HTTPS will be enforced via ACM and ALB. IAM roles following the principle of least privilege will govern all AWS service interactions. Input validation and sanitization will be implemented on both client and server to mitigate common vulnerabilities such as XSS and SQL/NoSQL injection.

5. Preliminary Findings on the Likely AWS Services to be Used

Based on preliminary research into the project requirements, the following AWS services have been identified as the most suitable offerings to support the deployment of EchoMateLite:

- **Amazon EC2 (Elastic Compute Cloud):** EC2 will host the Node.js/Express.js backend server. A t2.micro instance (AWS Free Tier eligible) will be provisioned for the initial deployment. EC2 was chosen for its flexibility, full OS-level control, and suitability for hosting Node.js applications.
- **Amazon S3 (Simple Storage Service):** S3 will serve a dual purpose: storing user-uploaded profile images as objects, and hosting the static build files of the React.js frontend. S3 is chosen for its virtually unlimited storage, high durability (99.999999999%), and cost-effectiveness.
- **Amazon CloudFront:** CloudFront will act as a CDN in front of the S3-hosted React frontend, improving global delivery speeds and enforcing HTTPS for the static site.

- **Amazon Cognito:** Amazon Cognito will manage the user authentication lifecycle — sign-up, sign-in, and token-based session management — through user pools, reducing the overhead of building a custom authentication system.
- **Amazon RDS / MongoDB Atlas:** The database layer will be implemented using MongoDB (via MongoDB Atlas or Amazon DocumentDB) for its JSON-document model, which is natively compatible with the MERN stack and well-suited for storing user profiles and posts.
- **Elastic Load Balancer (ELB) + AWS Certificate Manager (ACM):** An Application Load Balancer will distribute incoming traffic to the EC2 backend, while ACM provides a free SSL/TLS certificate to enable HTTPS for the API and the deployed application.
- **AWS IAM (Identity and Access Management):** IAM roles and policies will enforce the principle of least privilege across all AWS services, ensuring that each component (EC2, S3, etc.) only has the permissions strictly necessary for its function.

These services collectively form a robust, scalable, and cost-efficient cloud architecture for EchoMateLite. The use of AWS Free Tier resources (EC2 t2.micro, S3, Cognito, etc.) ensures the project remains within budget during the development and evaluation phases, while also providing a realistic, production-grade deployment experience.